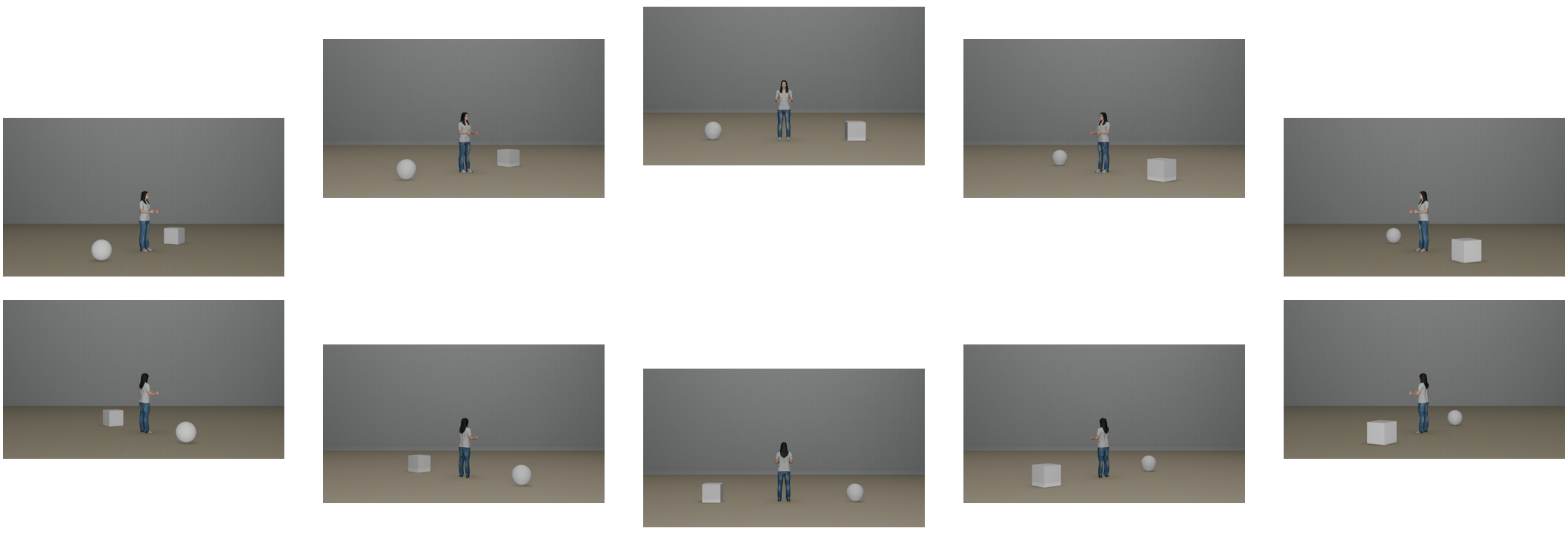
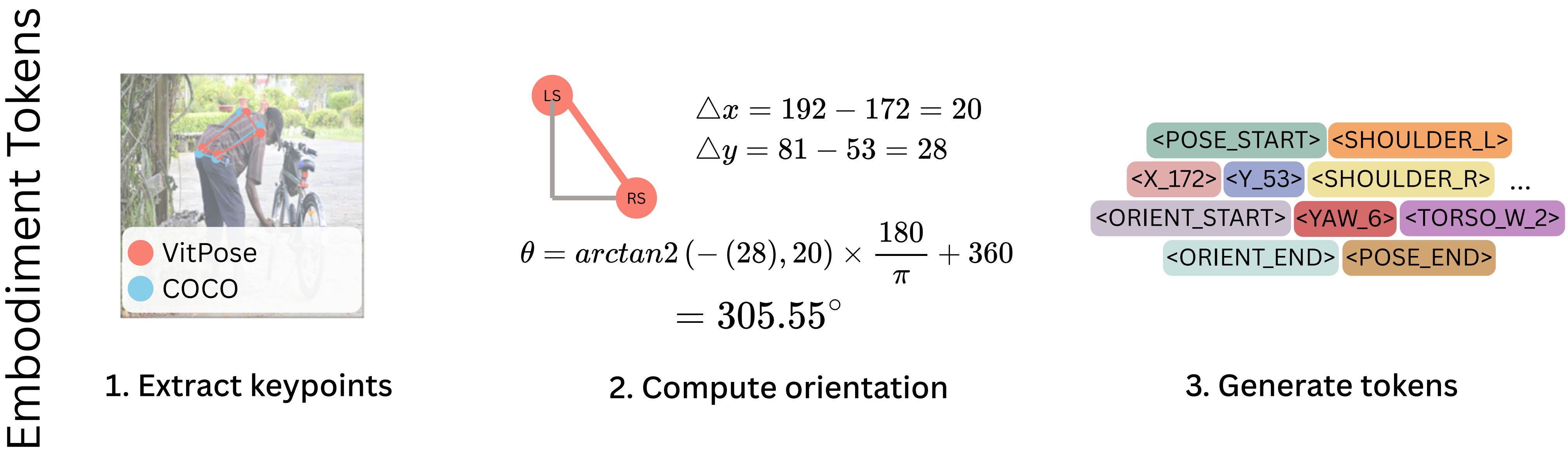
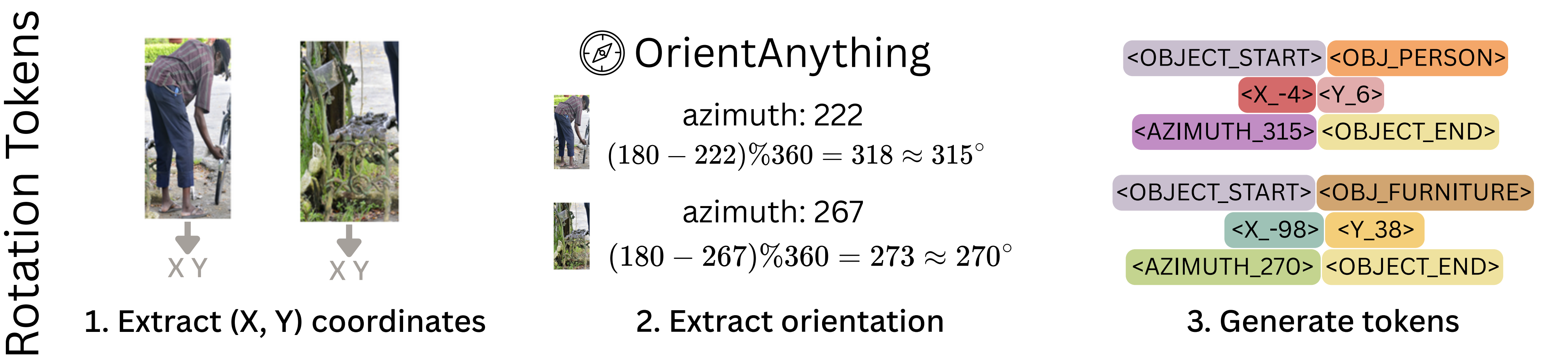
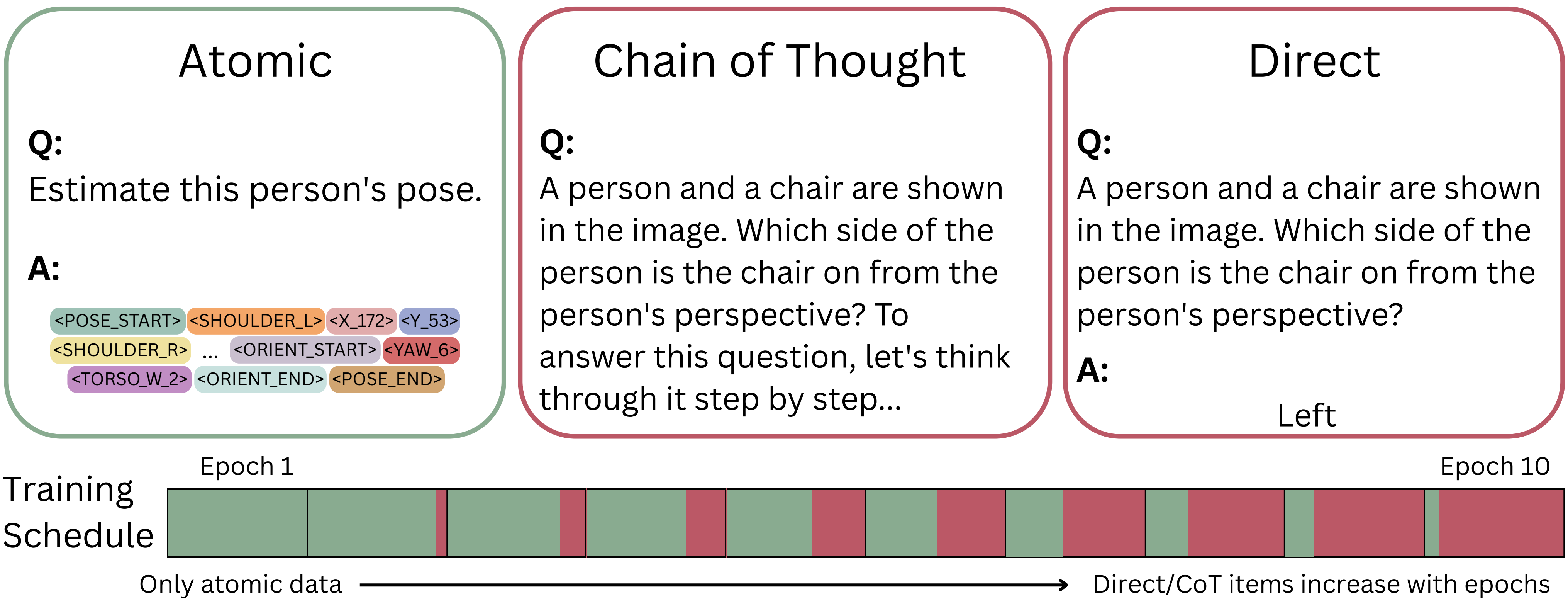
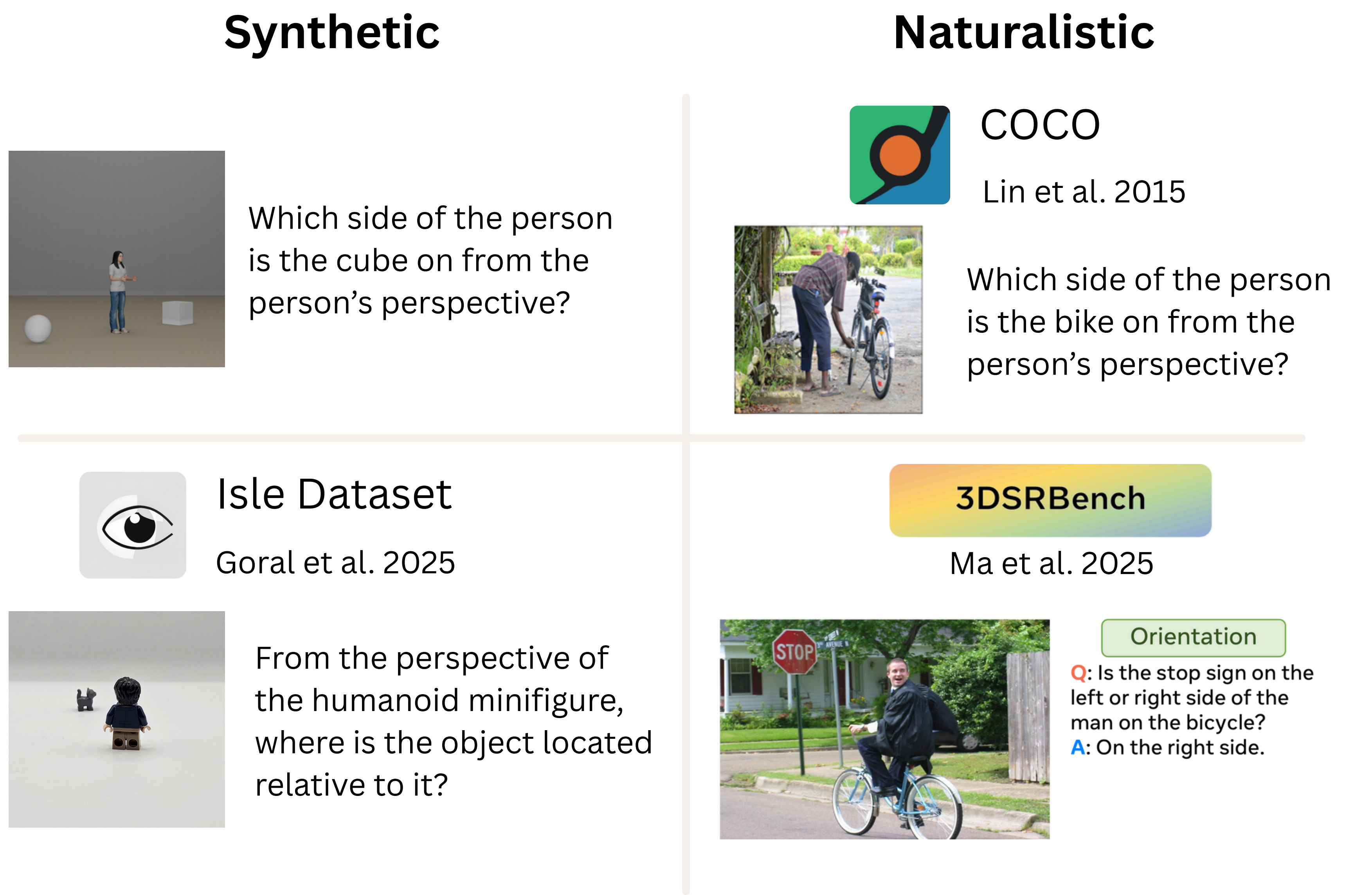



Embodiment Tokens
VitPose
COCO
1. Extract keypoints
LS
RS
$\triangle x = 192 - 172 = 20$
$\triangle y = 81 - 53 = 28$
$\theta = arctan2\,(-(28), 20) \times \frac{180}{\pi} + 360$
$= 305.55^\circ$
2. Compute orientation
<POSE_START> <SHOULDER_L>
<X_172> <Y_53> <SHOULDER_R> ...
<ORIENT_START> <YAW_6> <TORSO_W_2>
<ORIENT_END> <POSE_END>
3. Generate tokens


Rotation Tokens
X Y
X Y
1. Extract (X, Y) coordinates
OrientAnything
azimuth: 222
$(180 - 222)\%360 = 318 \approx 315^\circ$
azimuth: 267
$(180 - 267)\%360 = 273 \approx 270^\circ$
2. Extract orientation
<OBJECT_START> <OBJ_PERSON>
<X_-4> <Y_6>
<AZIMUTH_315> <OBJECT_END>
<OBJECT_START> <OBJ_FURNITURE>
<X_-98> <Y_38>
<AZIMUTH_270> <OBJECT_END>
3. Generate tokens


Atomic
Q:
Estimate this person's pose.
A:
<POSE_START> <SHOULDER_L> <X_172> <Y_53>
<SHOULDER_R> ... <ORIENT_START> <YAW_6>
<TORSO_W_2> <ORIENT_END> <POSE_END>
Chain of Thought
Q:
A person and a chair are shown in the image. Which side of the person is the chair on from the person's perspective? To answer this question, let's think through it step by step...
Direct
Q:
A person and a chair are shown in the image. Which side of the person is the chair on from the person's perspective?
A:
Left
Training Schedule
Epoch 1
Epoch 10
Only atomic data
Direct/CoT items increase with epochs


Synthetic
Which side of the person is the cube on from the person's perspective?
Naturalistic
COCO
Lin et al. 2015
Which side of the person is the bike on from the person's perspective?
Isle Dataset
Goral et al. 2025
From the perspective of the humanoid minifigure, where is the object located relative to it?
3DSRBench
Ma et al. 2025
STOP
Orientation
Q: Is the stop sign on the left or right side of the man on the bicycle?
A: On the right side.